NOVA-KV — Part D: Benchmarking Everything Together

Campaign `20261013T090846-615f4357f015-all`, measured at commit `615f435`. Every number below is the **median of 5 independent processes, with [min–max]** where given, and comes from a file under `results/part_d/`; the complete tables are `results/part_d/summary/summary.md` and the CSVs next to it. Nothing was estimated and no run was dropped. 300 measured runs, 0 failures, 0 problems found by the completeness check.

1. Environment and method

Machine. AMD Ryzen AI 7 350, 8 cores / 16 logical CPUs (cores 0, 2, 4, 6 reach 5.09 GHz; cores 1, 3, 5, 7 reach 3.51 GHz; CPU N+8 is the SMT sibling of N), L3 16 MiB, 30 GiB RAM. Ubuntu 26.04, Linux 7.0.0-28, GCC 15.2.0, Release build `-O3 -g -DNDEBUG -std=c++17 -fno-omit-frame-pointer`. Mains power, `performance` governor and EPP on all CPUs at each of 56 checkpoints (start, after every repetition, end); `kernel.perf_event_paranoid = 1`. The machine was 99 % idle when the campaign started. The battery was charging throughout and the ACPI temperature rose from 44 to 86 °C; the min–max ranges below are narrow, so no drift is visible, but thermal limiting cannot be excluded (`results/part_d/conditions.log`).

Placement. Reader (or joiner) thread i is pinned to the i-th of CPUs 0, 2, 4, 6, 1, 3, 5, 7: one logical CPU per physical core, fast cores first. The writer is pinned to CPU 15. The store's cleaner thread is not pinned (the store has no interface for it).

Two tracks, never in one table or figure.

- **Track R — supplied generator, preload-only.** The four drivers linked with the unmodified `supplied/workload_gen.cpp`, default parameters. Its writer loads all trades and *finishes before any reader starts*: there is no writer in these numbers. 20 % of its reads use `rng() % (m·n)` while trade ids are `i·100000 + j`, so only 100 of those 100,000 possible keys exist: measured hit rate **80.02 %**.
- **Track C — writer active.** `benchmarks/concurrent_bench.cpp`: the same schema, hot set and 80/20 mix, every requested key present, and one writer thread overwriting existing trades for the whole measured window (0.5 s warm-up, 5 s measured). Every read is validated; the writer's successful updates are counted. Two writer modes: paced at 10,000 updates/s, and unthrottled.

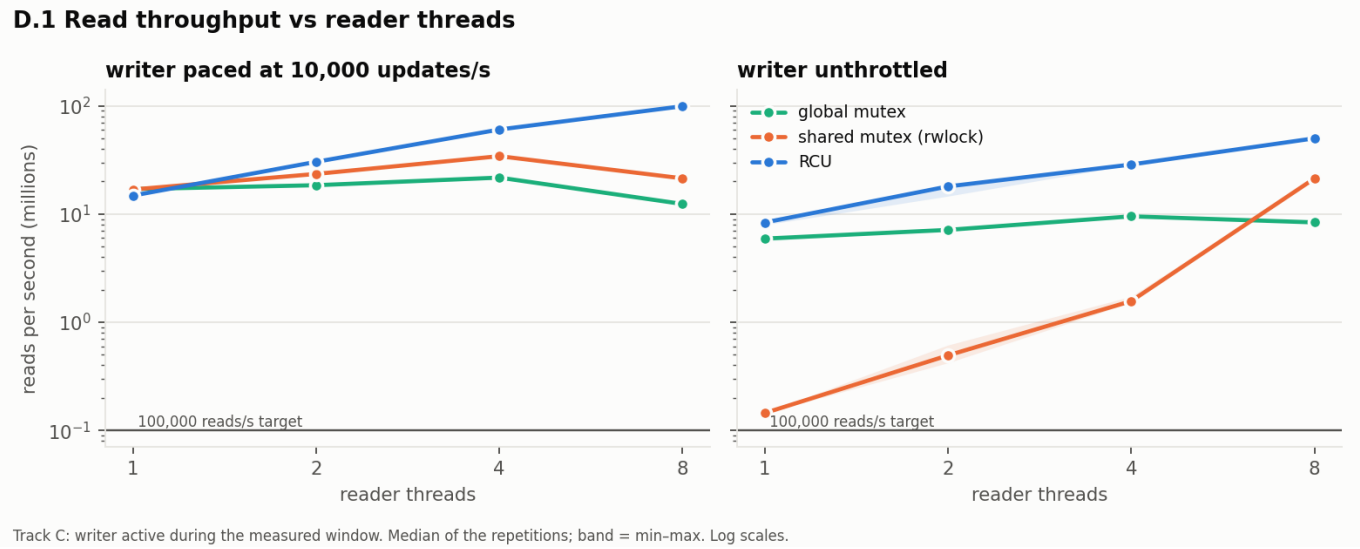
D.1 asks for the generator "with one background writer thread active throughout". The supplied generator cannot do that, so both tracks are reported and labelled.

Caveats that apply to all throughput numbers. Both harnesses time every `get` with two clock reads; in the profile of the RCU store with no writer those clock reads are 33 % of all cycles (`summary/profile_breakdown.txt`). The numbers are therefore lower bounds on what the stores can do, equally for all four. The generator prints latencies with one decimal of a microsecond.

Comparing RCU with the lock variants compares two whole designs. The Part A stores update one slot in place under a lock; the RCU store copies the 24 KiB segment it touches and a 2 KiB directory for every update and reclaims the old ones later. The difference between them is not the cost of a synchronization primitive alone.

2. D.1 — point-query throughput and the 100,000 RPS target

2.1 Track C: one writer active throughout



Track C: writer active during the measured window. Median of the repetitions; band = min–max. Log scales.

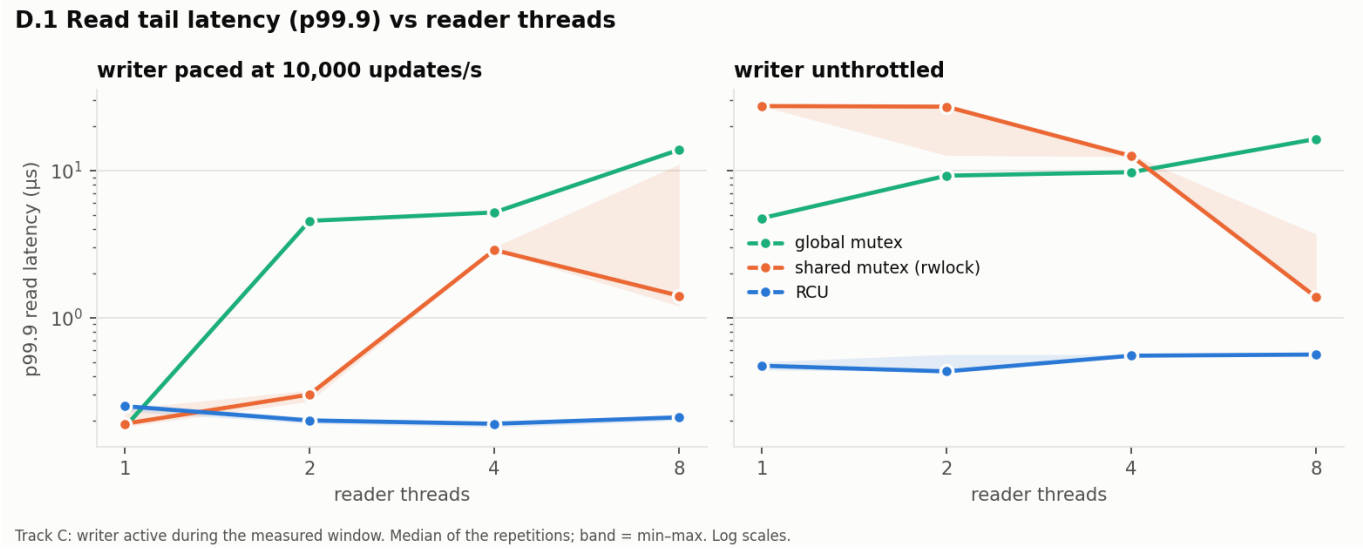
Reads per second (millions), median [min–max]:

Readers	global mutex, paced	shared mutex, paced	RCU, paced	global mutex, unthrottled	shared mutex, unthrottled	RCU, unthrottled
1	17.00 [16.98–17.20]	16.86 [16.78–16.87]	14.81 [14.77–14.92]	5.93 [5.65–6.16]	0.14 [0.14–0.15]	8.36 [7.96–8.87]
2	18.47 [18.08–18.85]	23.49 [23.18–23.90]	30.39 [30.28–30.55]	7.14 [7.12–7.40]	0.49 [0.42–0.61]	18.00 [14.66–18.02]

Readers	global mutex, paced	shared mutex, paced	RCU, paced	global mutex, unthrottled	shared mutex, unthrottled	RCU, unthrottled
4	21.73 [20.98–22.15]	34.28 [34.14–34.42]	60.44 [60.23–60.79]	9.53 [9.45–9.59]	1.57 [1.52–1.73]	28.69 [28.55–28.90]
8	12.39 [12.35–12.46]	21.37 [21.15–21.67]	**99.25** **[98.81–99.54]**	8.37 [8.24–8.45]	21.30 [21.09–21.42]	**50.09** **[49.93–50.47]**

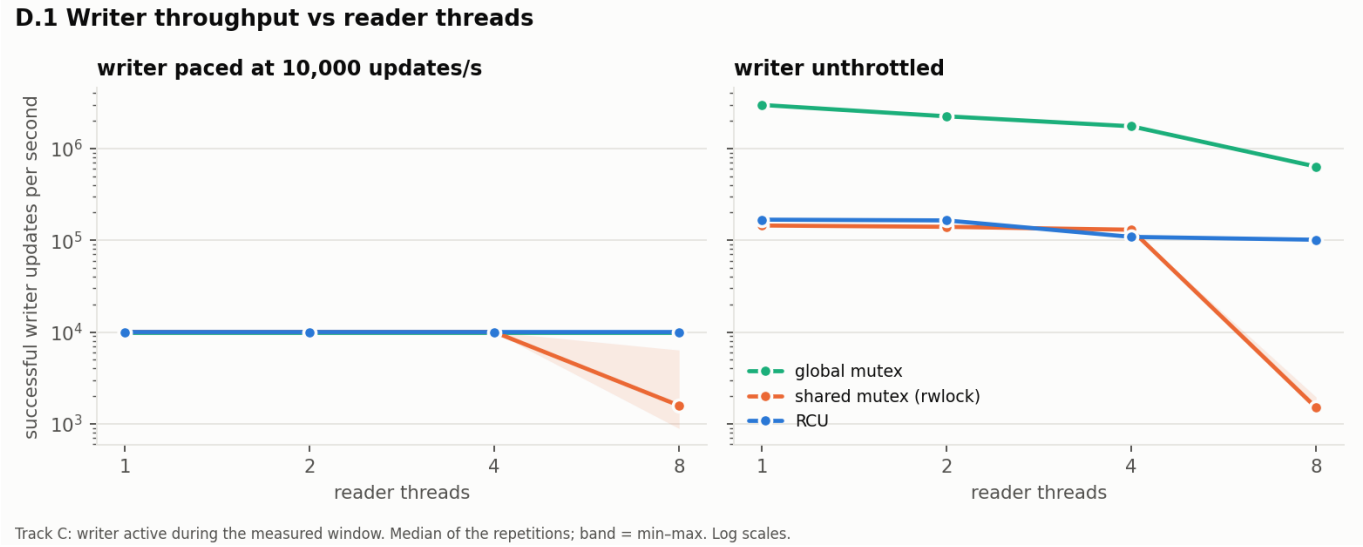
Read latency at 8 readers (µs):

Store, writer	p50	p99	p99.9	max, median run / worst run
global mutex, paced	0.110	7.616	13.824	166 / 992
shared mutex, paced	0.371	0.721	1.408	906 / 4,095
RCU, paced	0.050	0.141	0.210	87 / 143
global mutex, unthrottled	0.170	9.472	16.384	299 / 313
shared mutex, unthrottled	0.391	0.721	1.376	896 / 3,024
RCU, unthrottled	0.120	0.390	0.561	74 / 142



Successful writer updates per second:

Readers	global mutex, paced	shared mutex, paced	RCU, paced	global mutex, unthrottled	shared mutex, unthrottled	RCU, unthrottled
1	10,000	10,000	10,000	2,976,055	144,543	167,456
2	10,000	10,000	10,000	2,237,224	139,935	164,321
4	10,000	10,000	10,000	1,742,449	130,062	108,738
8	10,000	**1,582** **[890–6,351]**	10,000	633,443	**1,500** **[1,382–1,955]**	101,121



One thread only: the unsynchronized store reads 19.12 M/s alone and 18.21 M/s while doing one update per 100 reads on the same thread; it is never run with concurrent threads. The RCU store with no writer reads 16.67 M/s on one thread and 112.26 M/s on eight.

With the generator's own key distribution *and* a paced writer (8 readers): global mutex 13.40 M lookups/s, shared mutex 22.47 M, RCU 105.27 M, each with an 80.02 % hit rate (successful reads 10.72 / 17.98 / 84.24 M/s).

2.2 Where the lock-based stores stop scaling

- **Global mutex: after 4 readers.** Throughput rises from 17.0 to 21.7 M/s at 4 readers and *falls* to 12.4 M/s at 8 (paced writer); with the unthrottled writer 5.9 → 9.5 → 8.4 M/s. At 8 readers with the unthrottled writer the process makes 3.7 million context switches in a 5.5 s run (2.6 million with the paced one), 77 % of sampled on-CPU cycles are in the kernel (futex wait and wake) and another 13 % in the mutex functions; only 1.4 % are in the lookup itself

(`perf/flamegraphs/global_lock_r8_writer_max.svg`). Readers serialize on one lock and spend their time handing it over. p99 read latency grows from 0.07 µs at 1 reader to 7.6 µs at 8.

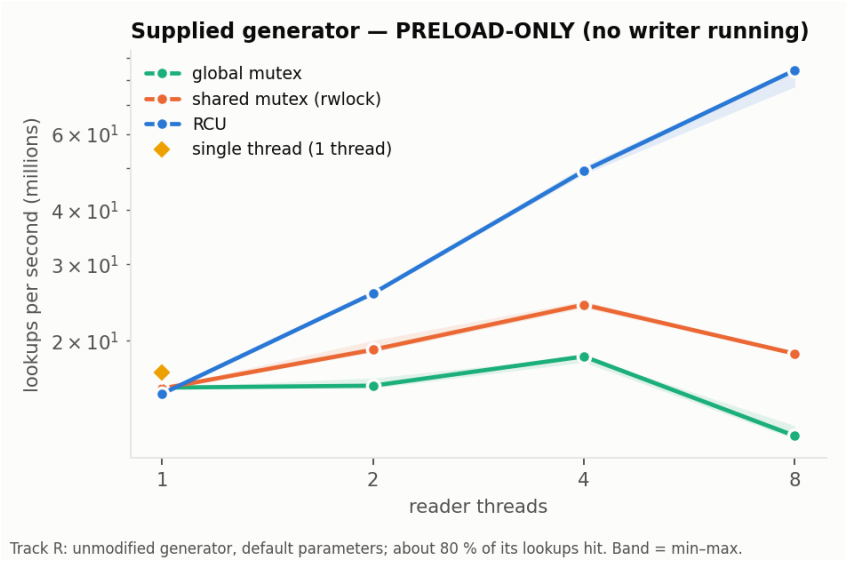
- **Shared mutex: after 4 readers with a paced writer** (16.9 → 23.5 → 34.3 → 21.4 M/s). Readers do not exclude each other, but every `get` writes the lock's reader count twice. At 8 readers 66 % of on-CPU cycles are in `pthread_rwlock_rdlock/unlock`, and `perf c2c` puts 67 % of all loads that hit a line modified by another core on **one cache line**, the lock itself (`perf/c2c/rwlock_r8_writer_max.c2c-report.txt`). **It also starves the writer:** asked for 10,000 updates/s it completes 1,582 at 8 readers, and unthrottled only 1,500. With an unthrottled writer and few readers the opposite happens: the writer holds the lock and reads collapse to 0.14 M/s at 1 reader; the "scaling" from 1.6 to 21.3 M/s between 4 and 8 readers is the readers taking the lock away from the writer, not parallel speed-up.
- **RCU: still rising at 8 readers** in both writer modes: 14.8 → 30.4 → 60.4 → 99.3 M/s paced (6.7× from 1 to 8 readers; the last four readers run on the slower cores and one shares its core with the writer), 8.4 → 18.0 → 28.7 → 50.1 M/s unthrottled.

2.3 The 100,000 reads/s target

RCU store, 8 readers	Median reads/s	Minimum of 5 runs	100,000/s
Track C, writer paced at 10,000/s	99,249,401	98,806,210	**met**
Track C, writer unthrottled (101,121 updates/s)	50,088,105	49,934,831	**met**
Track R, supplied generator (preload-only)	84,185,176 lookups/s	77,088,328	**met**

The target is met by a factor of about 500 to 1000 in every configuration measured, with p99.9 read latency of 0.21 µs (paced writer) and 0.56 µs (unthrottled). Because it is met, the assignment's fallback ("report the number you achieved and explain the remaining bottleneck, backed by a flame graph") does not apply; the flame graphs were taken anyway and are used in §2.5.

2.4 Track R — supplied generator, preload-only



Track R: unmodified generator, default parameters; about 80 % of its lookups hit. Band = min–max.

Readers	single thread	global mutex	shared mutex	RCU
1	16.89 [16.80–16.95]	15.58 [15.47–15.60]	15.50 [15.46–15.57]	15.07 [15.04–15.10]
2	—	15.74 [15.53–16.39]	19.05 [18.80–20.01]	25.69 [25.66–25.77]
4	—	18.39 [17.84–18.39]	24.20 [23.81–24.68]	49.29 [48.43–50.57]
8	—	12.05 [11.84–12.72]	18.66 [18.55–18.70]	84.19 [77.09–84.65]

Lookups per second (millions), **no writer running**, 80.02 % of lookups hit. The same shape as track C: the mutex stores peak at 4 readers, RCU keeps rising. p99.9 at 8 readers as printed by the generator: 7.7 µs (global mutex), 1.1 µs (shared mutex), 0.2 µs (RCU).

2.5 What RCU costs, and what limits it

Memory and reclamation. Every update copies 25,974 bytes (one 24 KiB segment and the 2 KiB directory) for a 32-byte value. With the paced writer the store holds 11 MiB at the end of every run and the backlog is empty. **With the unthrottled writer the cleaner does not keep up beyond 2 readers:**

Readers (unthrottled writer)	Memory held at end, MiB	Retired versions not yet reclaimed at end	Cleaner CPU, s of 5.5
1	31 [31–32]	0	4.73
2	31 [31–98]	0	4.87
4	655 [536–817]	23,852	5.42
8	3,662 [3,326–4,813]	138,157	5.22

The cleaner thread is fully busy at every reader count. At 8 readers the writer retires about 101,000 versions per second and the cleaner destroys about 87,000, so the backlog grows by roughly 25,000 versions (0.65 GiB) per second for as long as the writer runs flat out. The store has no back-pressure (a deliberate choice, decision D-040), so this ends only when the writer slows down or memory runs out. It drains completely afterwards (validated in every run). The audited Part B binary behaves the same when run alternately with the campaign's binary (75,000–100,000 unreclaimed versions after 3.5 s for both; a diagnostic outside the campaign, `results/part_d_diagnostics/part_b_reference_backlog.txt`), so it is not something Part C introduced. **A sustained unthrottled writer with 8 readers is not a steady state for this store on this machine.** The 10,000

updates/s workload is.

Reads slow down by half next to an unthrottled writer (99.3 → 50.1 M/s at 8 readers; 16.7 → 8.4 M/s for a single reader with idle cores all around it). What the evidence says:

- It is not lock or scheduler contention: readers take no lock, and the slowdown is uniform across reader threads (slowest/fastest 4.94/7.25 M/s against 11.48/16.58 M/s with no writer) and present with a single reader.
- The number of instructions per read hardly changes (447 → 541); cycles per read go from 335 to 928 and instructions per cycle from 1.33 to 0.58. The readers are stalled, not doing more work.
- `perf c2c` (AMD IBS load sampling): of the loads with a known data source, those served from L3 rise from 1.2 % to 3.6 % and those that hit a line modified by another core from 0.5 % to 2.9 %.
- **The hypothesis that writer-updated segment owner counts share cache lines with reader data (audit item R8) is only weakly supported.** Of 804 such "modified elsewhere" load samples at 8 readers, 58 % were taken by the writer's own commit and 24 % by the functions taking and releasing references (cleaner and writer): that is the writer and the cleaner contending with each other on owner and reference counts. **6.7 % (54 samples, on 46 different cache lines) were taken by the reader's lookup**, which is 0.07 % of all sampled loads. Readers do touch lines the writer or cleaner has just modified, but too rarely for that to explain a factor of two.
- What fits all of the above, without being proven by it: copy-on-write *relocates* what readers read. Each update publishes a new 24 KiB segment at a new address, so the copy of that partition a reader has in its own caches is useless and the next probe has to fetch lines from L3 or from the writer's core; at 100,000 updates/s that is 2.6 GB/s of freshly written index flowing through a 16 MiB shared L3 that also has to hold the readers' 9 MiB working set.
- The profile cannot separate this further. In the flame graph the reader threads' cycles shift from the lookup (64 % → 17 %) to the harness's clock reads (33 % → 49 %); a timestamp read waits for the loads before it, so stall time is charged to it. That shift is an artefact of attribution, not evidence that reading the clock became slower.

No optimization was attempted. A test that would decide it: per-thread hardware counters for the reader threads alone, and the same workload against a store that updates in place.

perf stat for the three stores at 8 readers with the unthrottled writer: CPUs utilized 7.30 / 7.90 / 8.79 (global mutex / shared mutex / RCU: the RCU process also runs its cleaner), context switches per second 674,661 / 13,805 / 27,545, peak resident memory 39 / 14 / 4,147 MiB.

3. D.2 — reclamation-batch tuning

`set_reclaim_batch(n)`: one reclamation pass frees at most n retired versions. Settings 1, 8, 64, 256, 1024; reclamation disabled; and the naive stop-the-world sweep (all readers frozen by a lock, the whole retire list freed at once). RCU store, 8 readers, unthrottled writer, 3 s windows (1 s with reclamation disabled, which retains everything). In every epoch-based run the largest number of versions freed in one pass equalled the batch (1, 8, 64, 256, 1,024) and never exceeded it.

Two workloads, **not comparable with each other**:

- **A, steady point gets:** the D.1 benchmark (100,000 trades of 32 bytes).
- **B, backlog-building:** `benchmarks/reclaim_bench.cpp`, 2,000 keys of 4 KiB, and a thread that holds a Snapshot for 20 ms out of every 50 ms. While it is held nothing retired after it can be freed, so thousands of versions accumulate and are released in passes. At the end the writer stops, the last snapshot is released, and the time to drain is measured with the readers still running.

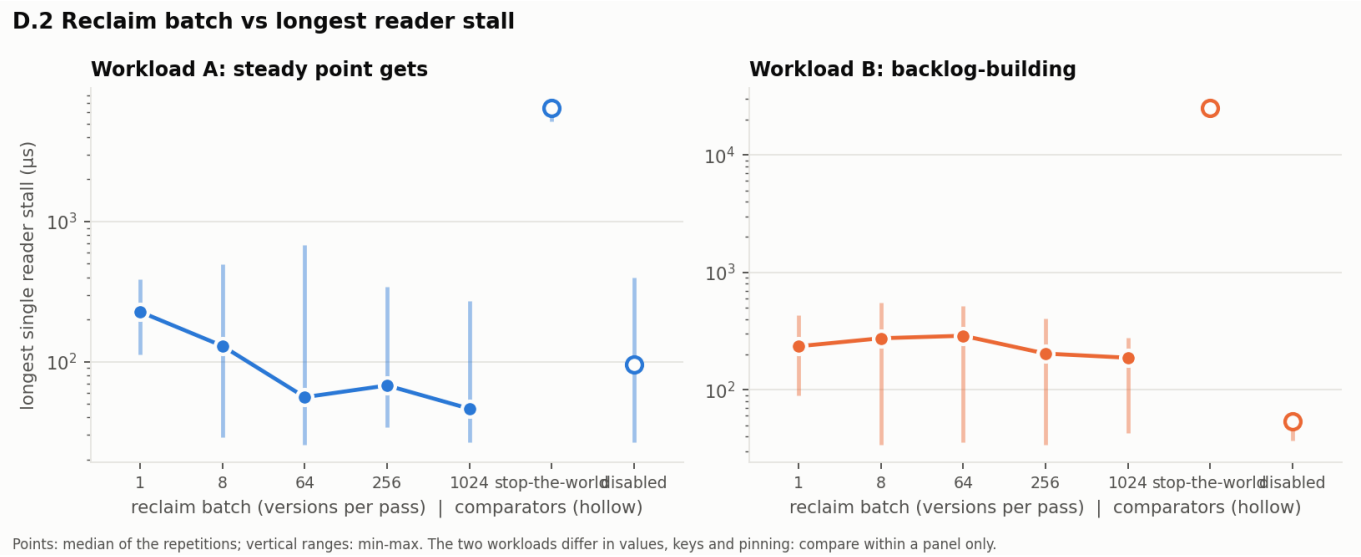
3.1 Results

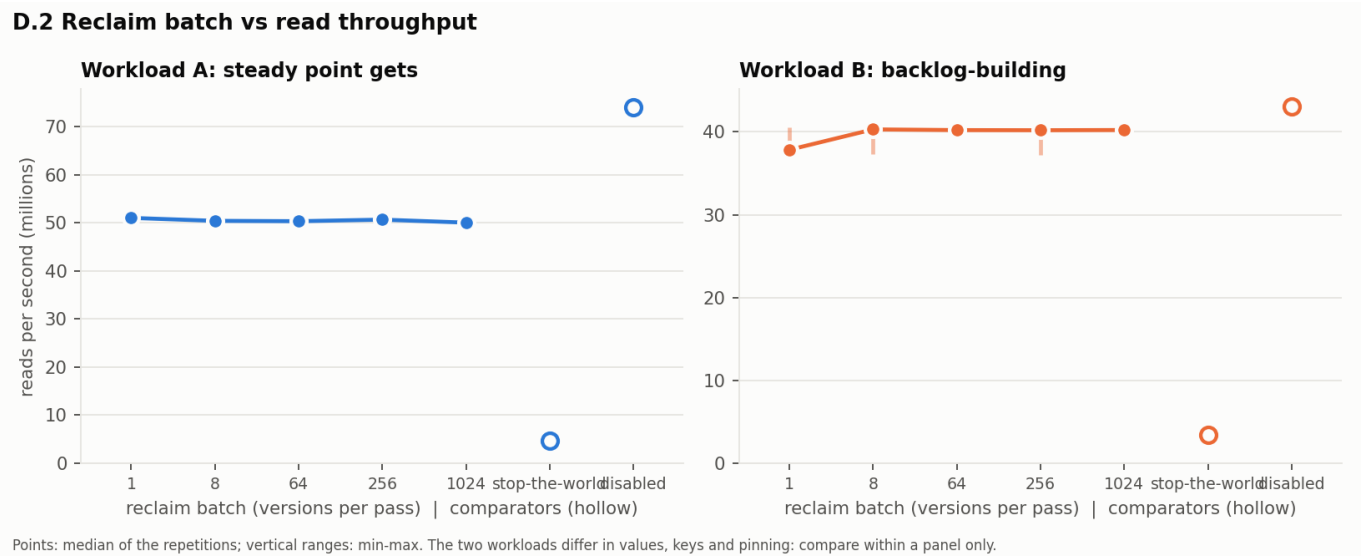
Workload A:

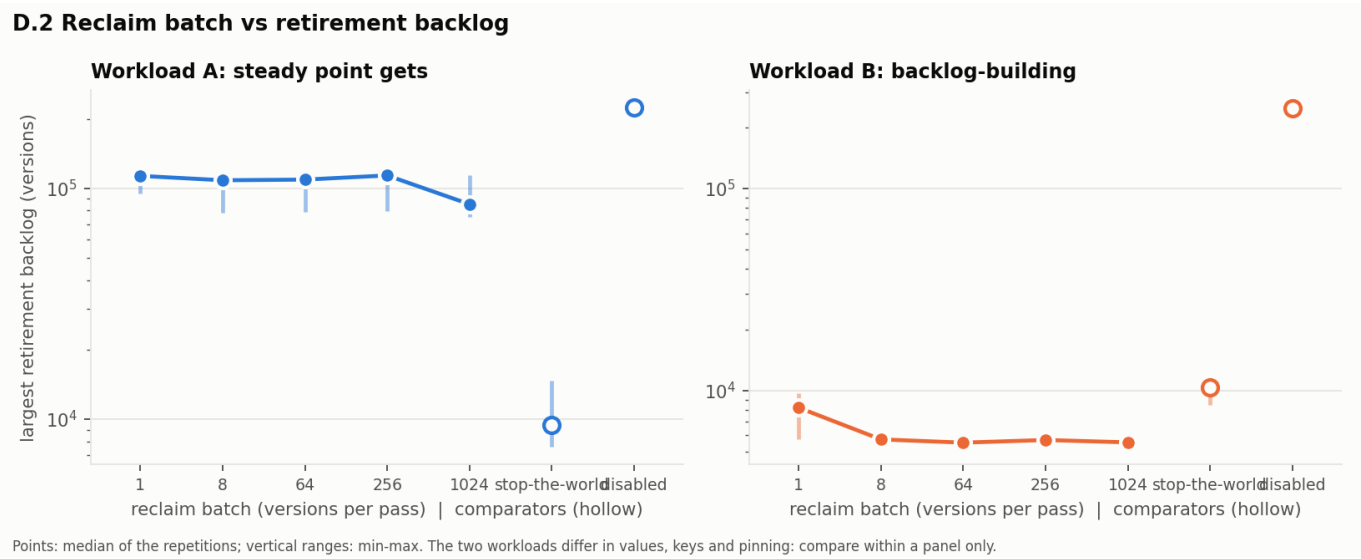
Setting	Reads/s (M)	Longest single reader stall, µs: median run / worst run	p99.9 µs	Versions reclaimed/s	Backlog max	Memory at end, MiB	Passes
batch 1	50.99 [50.53–51.14]	227 / 388	0.551	81,672	113,162	2,989	245,037
batch 8	50.38 [49.87–50.77]	129 / 495	0.561	87,068	108,314	2,862	32,730
batch 64	50.31 [50.12–50.52]	56 / 683	0.561	87,419	109,026	2,878	4,193
batch 256	50.63 [49.89–50.80]	68 / 340	0.561	83,175	113,650	3,002	1,076
batch 1024	50.01 [49.85–50.56]	46 / 272	0.561	97,374	85,273	2,257	732
disabled (1 s)	73.97 [73.84–74.26]	96 / 396	0.451	0	224,171	4,405	—
stop-the-world	4.62 [3.93–5.05]	**6,436 / 6,777**	2.144	193,026	9,429	30	—

Workload B:

Setting	Reads/s (M)	Longest single reader stall, µs: median run / worst run	p99.9 µs	Backlog max	Memory max, MiB	Final cleanup: backlog → ms to drain
batch 1	37.84 [37.66–40.57]	235 / 430	0.481	8,230	83.7	8,133 → 29.9
batch 8	40.30 [37.36–40.59]	275 / 550	0.481	5,730	66.7	5,760 → 18.1
batch 64	40.21 [40.10–40.41]	289 / 518	0.481	5,522	64.6	5,553 → 18.4
batch 256	40.20 [37.19–40.35]	203 / 406	0.481	5,691	65.3	5,734 → 15.3
batch 1024	40.22 [40.07–40.34]	188 / 278	0.481	5,540	65.0	5,553 → 14.6
disabled (1 s)	43.09 [42.94–43.17]	54 / 62	0.451	250,128	1,725	—
stop-the-world	3.42 [3.26–3.46]	**25,061 / 26,578**	10.112	10,406	97.7	10,447 → 15.7







3.2 What the batch size trades off here

The assignment expects "a bigger batch should mean higher throughput but a longer worst-case stall". **That is not what was measured**, in either workload:

- **Read throughput does not depend on the batch.** 50.0–51.0 M/s in workload A and 37.8–40.3 M/s in workload B across a thousandfold range of batch sizes, with p99.9 identical to three digits. The cleaner works on its own thread and takes no lock a reader takes, so how it groups its work is invisible to `get`.
- **The longest reader stall does not grow with the batch.** Medians of 227, 129, 56, 68 and 46 µs for batches 1 to 1024 in workload A, and 235, 275, 289, 203, 188 µs in workload B; the worst single run was at batch 64 and batch 8 respectively. These maxima are single events in about 150 million reads. With reclamation **disabled** the longest stall is of the same order (96 µs median, 396 µs worst run in workload A), so the stalls that occur are not caused by reclamation passes; with no writer

at all the maximum is 25 µs (85 µs worst run). There is no evidence that a reclamation pass stalls a reader, at any batch size.

- **What the batch does change is how fast versions are reclaimed, i.e. memory.** A pass has a fixed overhead (one attempt to advance the epoch, a scan of the reader slots), so one version per pass is the least efficient: in workload B, batch 1 leaves a backlog about 45 % larger (8,230 against about 5,600 versions), 28 % more memory, and takes twice as long to drain (29.9 ms against 14.6 ms). In workload A, where the cleaner is the bottleneck, batch 1024 reclaims 97,374 versions/s against 81,672 for batch 1 and ends with 25 % less memory. Between 8 and 256 the differences are within the run-to-run range.

So for this design the tunable trades **reclamation throughput and memory** against the granularity of the cleaner's work, not reader throughput against reader pauses. A larger batch is better or equal on every measure taken here; the reason to keep it bounded is that a pass is the unit in which the cleaner re-checks the epoch, and the bound is what makes its work between two checks finite.

3.3 The two reference points

- **Reclamation disabled** reads fastest (74.0 M/s in workload A against about 50) because nothing is being freed: no cleaner competing for a core and for the allocator. It retained 4.4 GiB in 1.5 s in workload A and 1.7 GiB in workload B. It is not a usable configuration.
- **Stop-the-world** is the fastest way to free memory (193,026 versions/s; 30 MiB held where the epoch-based runs hold 2–3 GiB) and lets the writer run faster (160,704 against about 101,000 updates/s), because the sweep has no batch limit and nothing else runs meanwhile. It pays with the readers: **read throughput falls to 4.6 M/s (one eleventh) and the longest reader stall rises to 6.4 ms in workload A and 25.1 ms in workload B.**

Bounded reclamation against stop-the-world, worst reader pause (median of the longest stall; the epoch figure is the worst batch size):

Workload	Stop-the-world	Epoch-based (worst batch)	Reduction
A, steady	6,436 µs	227 µs	28×
B, backlog-building	25,061 µs	289 µs	87×

4. D.3 — join and multi-key writes

4.1 Index-nested-loop join, writer active and paused

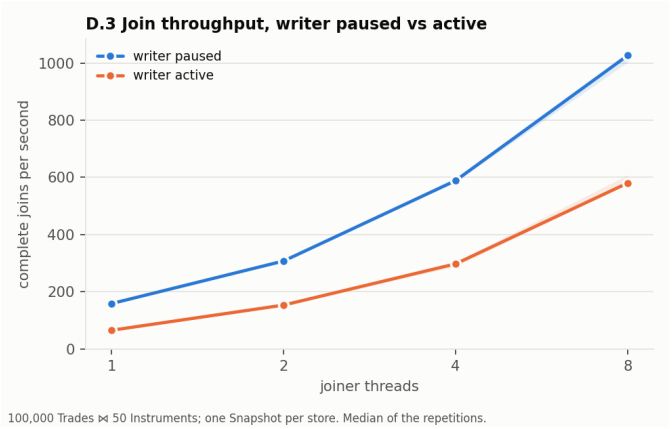
The join of Part C.2 (`benchmarks/join.h`): for every key that `keys()` returns for the Trades table, fetch the trade, probe the Instruments table's secondary index with its symbol, and fetch every matching instrument. 100,000 trades, 50 instruments, 100,000 joined rows per join; every join is validated (complete; pairs match; the inner table of one version).

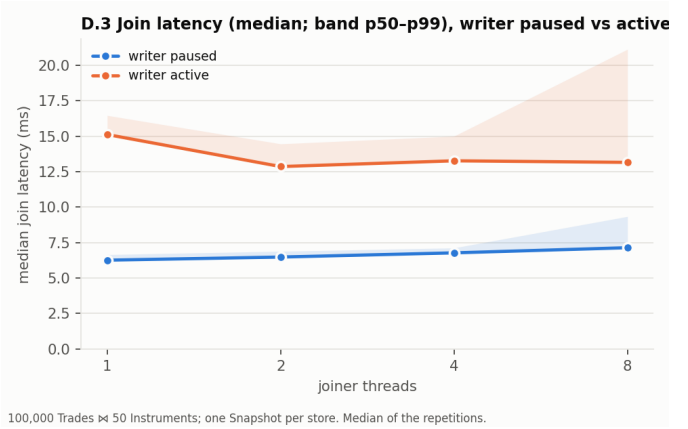
Snapshots: one per store. The join takes one Snapshot of Trades and one of Instruments at its start and uses each for every call on its own store. Each table is read from one immutable version for the whole join; the two versions are not guaranteed to be one atomic instant; and it is two handles, not the "exactly one Snapshot handle" of the assignment's wording (a recorded deviation, decision D-060).

When active, the writer issues single Trade puts that change the trade's symbol and, after every 64 of them, one transaction rewriting all 50 instruments.

Joiners	Writer	Joins/s	Joined rows/s (M)	p50 ms	p99 ms	p99.9 ms	max ms, worst run	Memory max, MiB	Backlog max
1	paused	159.0 [157.2–160.2]	15.90	6.26	6.65	11.31	11.80	9.1	0
1	active	65.6 [65.1–65.9]	6.56	15.12	16.46	16.75	16.87	149.1	5,431
2	paused	307.8 [304.1–311.0]	30.78	6.48	6.89	10.45	11.02	9.1	0
2	active	153.6 [150.4–154.9]	15.36	12.86	14.45	14.63	14.72	129.8	4,630
4	paused	588.7 [584.0–598.6]	58.87	6.78	7.12	10.57	10.64	9.1	0
4	active	297.8 [289.2–303.9]	29.78	13.26	15.00	15.15	16.14	127.7	4,547
8	**paused**	**1,026 [1,003–1,036]**	102.62	**7.14**	9.34	11.61	12.88	9.1	0
8	**active**	**580.3 [576.1–608.0]**	58.03	**13.15**	21.12	24.08	24.64	446.6	16,926

Writer while the joins ran: 163,318 Trade puts/s and 2,552 Instrument transactions/s with one joiner; 108,352 and 1,693 with eight.





What the concurrent writer costs a join. At 8 joiners: 1.84× the median latency (7.14 → 13.15 ms), 2.3× the p99 (9.34 → 21.12 ms) and 43 % of the throughput (1,026 → 580 joins/s). The same factor of about two as for point reads next to an unthrottled writer (§2.5), and for the same reason: the join never waits for the writer, but the rows and index segments it reads keep being republished at new addresses. A join is never blocked and never returns a wrong or partial result in either mode.

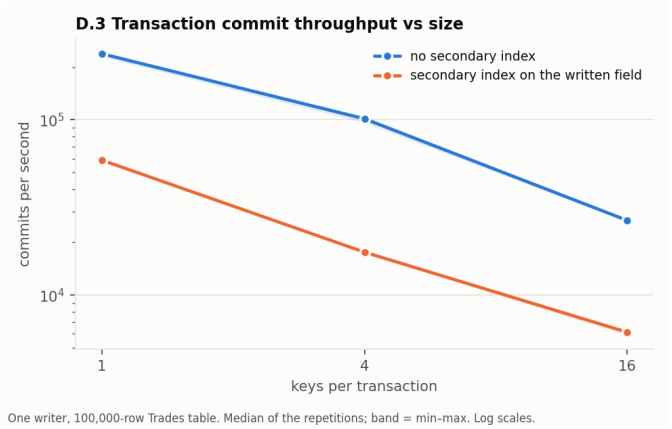
Joins hold reclamation back. A join pins both stores for its 7–25 ms, so everything retired meanwhile waits: up to 447 MiB held and 16,926 unreclaimed versions with 8 joiners and the writer active, against 9.1 MiB with the writer paused. All of it is reclaimed when the joins end.

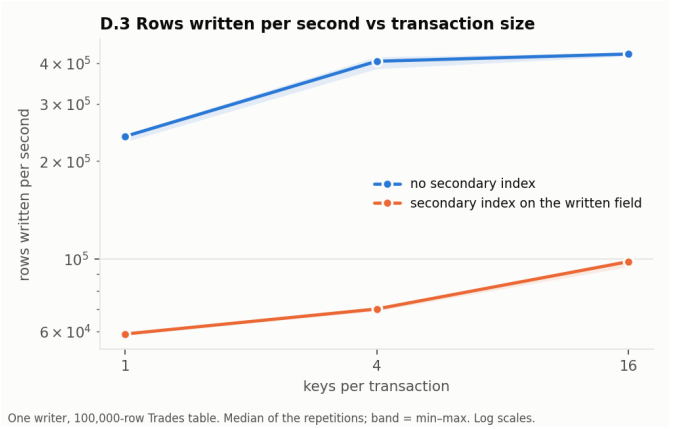
The generator's `use_join` path is not this join. It follows each successful `get` with one `index_get` on the Instruments store and discards the result. On the RCU store, 8 readers, preload-only: 61.30 M operations/s against 81.48 M without the probe. Its probe passes an eighth symbol byte that the generator never writes; in these binaries that byte was 0 in all 100,000 stored trades and the probe matched **100 %** (measured from the store after each run).

4.2 Transaction commit throughput by size

One writer committing transactions of N Trades into the 100,000-row table, keys spread over the table (so N keys fall into N different hash partitions almost always). "No index" is the Trades table as the assignment loads it; "indexed" adds a secondary index on the field every write changes, the worst case for index maintenance.

Keys	Index	Commits/s	Rows/s	p50 µs	p99 µs	p99.9 µs	max µs, worst	Segments cloned per commit	Bytes copied per commit	Bytes copied per row
1	none	238,103 [229,076–240,839]	238,103	3.7	8.7	12.0	165	1.00	26,102	26,102
4	none	101,298 [96,078–104,522]	405,192	9.2	18.6	24.2	103	3.98	97,729	24,432
16	none	26,655 [26,286–26,867]	426,486	37.5	69.4	83.8	145	15.55	376,025	23,502
1	indexed	58,824 [58,446–59,059]	58,824	16.3	29.6	34.9	90	1.00	140,817	140,817
4	indexed	17,555 [17,304–17,584]	70,219	57.0	76.2	85.8	380	3.98	523,721	130,930
16	indexed	6,134 [5,918–6,220]	98,150	161.6	207.8	232.1	338	15.55	1,472,800	92,050





- **Commit throughput falls almost in proportion to the transaction size**: 238,103 → 101,298 → 26,655 commits/s for 1, 4, 16 keys, because each key lands in a different partition and each touched partition is one 24 KiB segment to copy. A commit costs about 24 KiB of copying per key plus a fixed 2 KiB directory and about 250 owner-count updates.
- **Rows per second rise** (238,103 → 405,192 → 426,486) because that fixed part and the single publication are shared by more rows; the gain is small since the per-key copy dominates.
- Each commit is still one publication, whatever its size.

- **A secondary index on the written field costs about a factor of four** (58,824 against 238,103 commits/s for one key): changing a row's indexed value rebuilds two of the 64 index partitions, about 115 KiB. With 100,000 rows and only 50 distinct values those partitions hold whole posting lists of 2,000 keys. This is the cost of copying index segments; the table the assignment indexes (Instruments, 50 rows) pays almost nothing, and the writer rewrote it in whole-table transactions 2,552 times per second during the join runs.

4.3 Where copy-on-write helps little, or hurts

Measured: the same Trade rows written as one transaction and as single puts.

Case	Publications	Elapsed ms	Rows/s	Segments cloned	Bytes copied	Bytes copied per row (payload 32)
100,000 rows, one transaction	1	13.40 [13.18–13.67]	7,462,767	256	6.1 MB	61
100,000 rows, single puts	100,000	559.41 [556.82–560.87]	178,759	100,000	2,610 MB	26,103
4,096 rows over all partitions, one transaction	1	0.90 [0.73–1.02]	4,526,992	256	6.1 MB	1,500
4,096 rows, single puts	4,096	23.94 [23.59–24.25]	171,083	4,096	107 MB	26,069
16 rows, one transaction	1	0.03	557,414	15	0.35 MB	21,632
16 rows, single puts	16	0.06	258,490	16	0.40 MB	25,088

Copy-on-write copies a whole 24 KiB segment to change one 24-byte slot: **816 bytes copied per byte of payload** for a single-key update. That is where it hurts:

1. **Small, frequent, scattered writes.** Every update pays the full segment; that is why the global-mutex store, which updates one slot in place, completes 2.98 million updates/s where the RCU store completes 167,456 (D.1, 1 reader, unthrottled), and why the cleaner cannot keep up at 8 readers (§2.5).
2. **A very large transaction that touches most of the table**, the assignment's example. It is efficient *per row* (61 bytes per row for 100,000 rows: the segments are copied once and filled completely), but: it copies the entire index (6.1 MB here, the whole table); the old and the new version coexist, so memory doubles until the old one is reclaimed, and stays doubled for as long as any reader or snapshot is in the old version; and it is one 13 ms-long commit during which no other write can be published. For a table larger than the caches this is a full rewrite to make a change that an in-place store would make row by row.
3. **A medium transaction spread thinly** is the worst per row among the batched cases: 4,096 rows over all 256 partitions copy the same 6.1 MB as 100,000 rows (1,500 bytes per row), because every partition is touched and each is copied whole whatever fraction of it changes.
4. **Workloads with long readers.** A join or any long-held snapshot keeps every version published during it alive (447 MiB during 13 ms joins here); a lock-based store has no such cost because it has no old versions.
5. **Indexes on low-cardinality fields of large tables** (§4.2).

Where it helps: reads never wait (p99.9 of 0.21–0.56 µs against 1.4–16 µs for the lock stores at 8 readers), read throughput scales with cores, the writer is not starved by readers (10,000 of 10,000 paced updates where the shared mutex delivers 1,582), and multi-row consistency for a reader costs nothing extra.

5. Failed or inconclusive hypotheses

- **"A bigger batch gives higher read throughput and a longer worst-case stall."** Not observed (§3.2). The batch affects reclamation throughput and memory.
- **"Owner counts share cache lines with reader data and cause the read slowdown next to a writer" (R8).** Only weakly supported: reader loads that hit a remotely modified line exist but are 0.07 % of sampled loads (§2.5). The dominant coherence traffic is between the writer and the cleaner. The slowdown itself is real and its cause is not established beyond "the readers are stalled on memory"; no change was made to the store.
- **"The steady workload is steady."** It is not at 8 readers with an unthrottled writer (§2.5): the backlog grows throughout.

6. Limitations and threats to validity

- One machine, one compiler, a laptop with two kinds of cores; the battery was charging and the temperature rose during the campaign. Ranges are narrow, but thermal effects are not excluded.
- Five repetitions of 3–5 s windows. "Longest stall" is a maximum over about 100 million samples and is sensitive to a single preemption; it is reported per run (median and worst), not averaged.
- Throughput includes two clock reads per operation (about a third of the cycles of an uncontended read).
- perf stat counters are per process, not per thread; they cannot isolate the readers. Kernel frames in the flame graphs have no symbols (`kptr_restrict = 1`).
- The cleaner thread is not pinned.
- `perf c2c` samples loads; a sample count is evidence of where contention is, not a measurement of its cost.
- Track R has no concurrent writer by construction and an 80 % hit rate; it is a compatibility reference, not the D.1 measurement.
- The join uses one Snapshot per store (two handles): each table is consistent, the pair is not an atomic cross-store snapshot.
- RCU and the lock stores differ in storage and update strategy, not only in synchronization.
- With reclamation disabled the window is 1 s; its throughput is not directly comparable with the 3 s runs.

7. Raw data

What	Where (under `results/part_d/`)
Machine, toolchain, build flags, executable hashes, prerequisites	`env/`
Conditions at 56 checkpoints	`conditions.log`
D.1 track C: one CSV row per run; copy-on-write and reclamation counters	`d1/concurrent/runs.csv`, `cow.csv`
Latency histograms, one per run (read and write series)	`d1/concurrent/hist/`, `d2/*/hist/`, `d3/*/hist/`
perf stat and peak RSS per run; command line and exit status per run	`*/perf_stat/`, `*/rss/`, `*/stdout/`
D.1 track R	`d1/reference/`
D.2 workloads A and B	`d2/steady/`, `d2/backlog/`
D.3 transactions, join, copy-on-write, generator paths	`d3/txn/`, `d3/join/`, `d3/cow/`, `d3/generator/`
Flame graphs (9 SVGs), folded stacks, compressed `perf script` output, commands	`perf/flamegraphs/`
perf c2c reports (4)	`perf/c2c/`
Summary tables, CSVs, profile regrouping	`summary/`
Figures (12)	`figures/`

Reproduce: `scripts/run_part_d.sh check`, then `scripts/run_part_d.sh all` (about 40 minutes; refuses to run without mains power, the `performance` governor, usable perf and an idle machine), then `scripts/aggregate_part_d.py results/part_d` and `scripts/part_d_profile_breakdown.py results/part_d`.